

R.O.B. VISION

VIRTUAL ROBOT COMPANION / MK.01



INSTALLATION & SETUP

One command per device, pairing, and first game check

VERSION 0.1.7

UNO Q · RETROPIE · BATOCERA

CHAPTER 01 / INSTALLATION & SETUP

Install and set up R.O.B. Vision

This guide gets a new Arduino UNO Q and one RetroPie or Batocera console ready to play. **Each device has one command for both installation and upgrades.** Run the same command again on that device whenever you want the latest release. R.O.B. Vision runs Buddy and the virtual game pieces on the UNO Q; your browser shows them. The console runs the NES game and sends its rendered light commands over your local network. You can pair more than one console, but only one game session is active at a time.

What you need

- An Arduino UNO Q with App Lab, plus a RetroPie or Batocera console. The installer checks Batocera for an installed FCEUmm or Nestopia core and a loadable wrapper. The package includes wrappers for x86_64, 32-bit x86, AArch64, ARMv7, ARMv6, and RISC-V 64; a native C compiler can build a wrapper if the packaged one cannot load. Live Batocera testing covered 43.1 x86_64.
- Both devices on the same trusted network. The UNO Q needs Python 3.9+; standard RetroPie needs Python 3.7+, gcc, openssl, sudo, systemd, uinput, and at least one installed NES core (lr-fceumm or lr-nestopia).
- curl, tar, Python 3, and sha256sum or shasum on the device running each installer command. The installer reports missing prerequisites.
- Your own legally obtained Gyromite or Stack-Up ROM. The console identifies the game by an exact registered NES ROM filename. Common World names are included; select **Edit ROMs** beside that console in Setup to add yours after pairing. See [game identification](#).

Before installing on RetroPie, exit any game **and EmulationStation**. Restarting the virtual joystick while EmulationStation is open can crash its input manager. On the UNO Q, stop another App Lab app before starting R.O.B. Vision.

1. Install or upgrade the UNO Q

Sign in as `arduino` and run this command in its terminal:

```
curl -fsSL https://github.com/mathan416/ROB-Vision/releases/latest/download/install.sh | sh -s -- -u -q
```

This is also the UNO Q upgrade command. Rerun it later to get the latest release; there is no separate upgrade command. The installer verifies the release package, installs and starts the App Lab app, and compiles and uploads the LED matrix sketch. It preserves pairing data during upgrades. It prints Mission and Setup links for both the UNO Q's `.local` name and available LAN IPv4 addresses. Open the printed **Setup** link in a browser on your laptop, tablet, or phone. A downloaded file:// copy of Setup is only a preview and cannot pair a live console.

2. Install or upgrade each console

Run this command **on RetroPie or Batocera**, replacing your `-u -q.local` with the UNO Q name or LAN IP printed by its installer:

```
curl -fsSL https://github.com/mathan416/ROB-Vision/releases/latest/download/install.sh | \
sh -s -- -c console your-uno-q.local
```

This is also the console upgrade command. Rerun it later on that console to get the latest release; it keeps the existing pairing. The same command detects the platform. It installs the game-frame integration, local receiver, launch and exit detection, and virtual Controller 2. On Batocera, it temporarily pauses and resumes the game menu

while updating the virtual joystick. On RetroPie, it requires EmulationStation to be closed. It does not install ROMs.

For a first install, leave the terminal open. The installer prints a **six-digit pairing code** and a **SHA-256 certificate fingerprint**, and waits up to five minutes. Do not enter a password or private key in Setup.

3. Pair in Setup

On the UNO Q Setup page, open **Console Link**. Enter the console address, the six-digit code, and the fingerprint exactly as printed. Choose **Pair Console**. Wait until the console terminal says **Pairing complete**, then select **Check Link** in Setup. The receiver and Controller Router start. Buddy is assigned to Player 2 automatically; existing physical-controller assignments stay in place.

Use **Pair Another Console** for a second machine. Paired consoles appear as separate rows. **Edit ROMs** opens that console's own filename registry. **Receiver Waiting** means this UNO Q has not heard from the console's R.O.B. Vision receiver recently; it does not mean the machine is powered off. A console uses one UNO Q at a time. Pairing it to a different UNO Q switches its receiver address; the old UNO Q retains a waiting row until you remove it. **Remove** and **Confirm Remove** revoke one console without unpairing the others. To reconnect a removed console, rerun its one-command install with `--pair` appended.

Review controller assignments

On the UNO Q Setup page, choose **Controllers** beside a paired console. Buddy stays on Player 2. The list shows physical controllers configured in EmulationStation and their current player assignments. On a new Router installation, known controllers are seeded in Player 1–4 order. Change an assignment if needed, exit the running game, and choose **Save Assignments**. **Test Inputs** listens briefly for buttons and directions. **Restore Previous** returns to the last saved assignment. RetroPie requires EmulationStation closed during first pairing or Router installation because new virtual controllers can upset its input manager.

4. Check a game

On RetroPie, launch the exact registered Gyromite or Stack-Up ROM with `lr-robvision-fceumm` or `lr-robvision-nestopia`. On supported Batocera, launching a registered ROM selects its R.O.B. Vision wrapper; an existing supported Nestopia choice can be retained. An ordinary NES core still plays the game but does not send Buddy commands.

Mission and Setup should show the game selected and **GAME FRAMES LINKED**. In Test mode, Buddy's red light blinks automatically and the UNO Q matrix shows **T**. In Gyromite Game A, use **Lower Blue** and **Lower Red** separately to check the gates, then **Release Both**. In Stack-Up Direct mode, move Hector onto a command key and watch Buddy and the blocks on Mission.

If the link is online but no game is selected, check the exact ROM filename and launch the game again. If the game is selected but frames are waiting, check the R.O.B. Vision core choice. The built-in Help and [Troubleshooting](#) give the next checks.

Upgrade or repair

Rerun that device's command from steps 1 or 2 to upgrade or repair it. The UNO Q app restarts while preserving pairing data; the console retains its credential and restarts its receiver. Exit an active RetroPie game and EmulationStation before reinstalling. The installer verifies the downloaded source package before changing installed files. Release packages also publish SHA256SUMS.

For advanced configuration and platform-specific service details, see the [Technical Reference](#), [RetroPie deployment](#), and [Batocera deployment](#).